

Trainings ▾

General Information

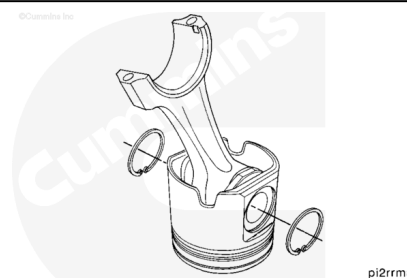
⚠ CAUTION ⚠

Articulated (two-piece) and non-articulated (one-piece) pistons are not able to be mixed within the same engine. Use of articulated and non-articulated pistons in the same engine can result in catastrophic engine damage.

Mechanically actuated injector engines were originally produced with articulated (two-piece) pistons. The articulated piston was later made obsolete by a non-articulated (one-piece) cast iron piston. If an articulated piston is to be replaced, all six pistons **must** be replaced with one-piece cast iron pistons or engine damage can result.

Preparatory Steps

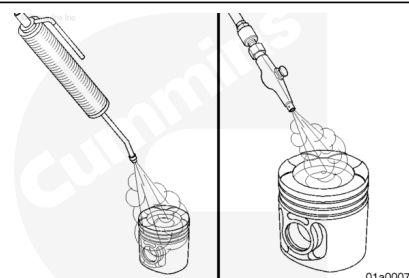
- Remove the piston from the connecting rod. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-054-tr.html)
- Remove and discard the piston rings. Refer to Procedure 001-047 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-047.html)



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious personal injury.



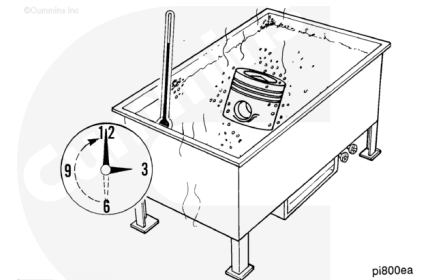
Use steam to clean the outer layers of carbon from the piston surface.

The bead blast method can be used to clean the top and crown of the piston. If this method is used, cover the ring grooves and skirt securely to prevent damage.

Note : It is only necessary to remove the carbon buildup. It is not necessary to make the piston crown appear new.

⚠ CAUTION ⚠

Do not contaminate the wash tank solution with bead blast materials. Use a kerosene emulsion-based solvent. Do not use a solvent that is higher than 9.5 pH. Do not use a solvent that contains chlorinated hydrocarbons with cresol, phenols, or cresylic components.

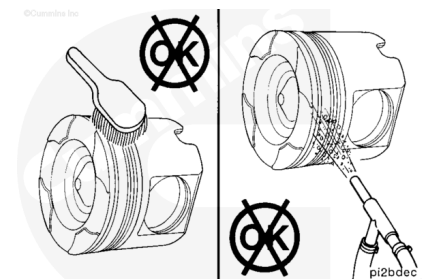


Allow the pistons to soak in the solvent for a minimum of 30 minutes.

Note : For best results soak the pistons several hours or overnight. Use a solvent that can be heated from 90°C to 95°C [194°F to 203°F]. Use a cleaning tank that will circulate and filter the solvent.

⚠ CAUTION ⚠

Do not use the bead blast method to clean the skirt. The piston skirt will be damaged by blast material being embedded in the aluminum.



⚠ CAUTION ⚠

Do not use a metal bristle brush to clean the pistons. Damage to the ring grooves will result.

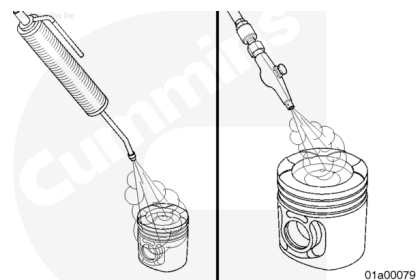
Use a brush to clean the ring grooves. Walnut shell blasting can be used on the piston crown. Repeat the soaking and scrubbing process until the piston is completely clean.

⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious personal injury.

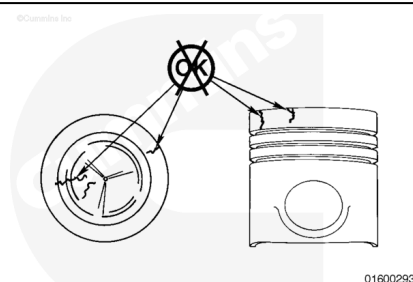
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

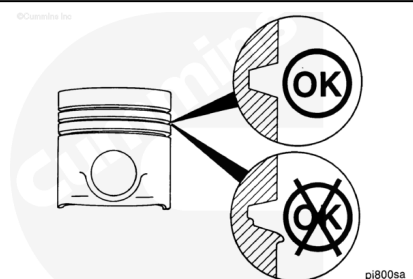


Use steam to rinse the solvent from the pistons. Dry the pistons with compressed air.

Inspect both the crown and skirt. Do **not** use the crown or skirt if cracks or scuff marks are found. Scratches in the skirt are **not** acceptable if they are visually **not** acceptable. Use Crack Detection Kit, Part Number 3375432, or equivalent, to check for cracks on the piston crown and in and around the piston pin bore. Do **not** use if cracked or scuffed. If the crown is **not** acceptable, replace the crown and pin.



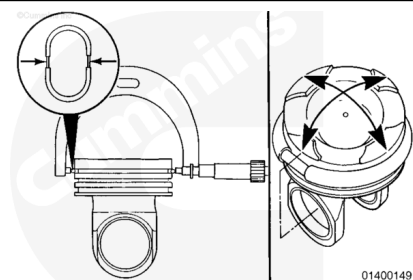
Inspect the ring grooves. The piston **must** be replaced if there is a visible ridge in the back of the groove. The piston **must** be replaced if a lip has formed on the outside diameter of the groove.



Use piston ring groove wear gauge, Part Number 3824846, and a 6 to 7 inch micrometer to inspect the top groove (compression ring).

Note : The piston **must** be replaced if measured dimensions are less than those in the table below.

Measure the top and second ring grooves (diameter over pins) when the piston temperature is at 21°C (70°F) in two places 90 degrees apart as shown.



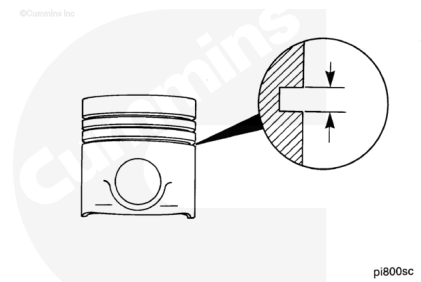
Ring Groove Wear Limits (Diameter Over Pins)

	mm		in
Top Ring	159.100	MIN	6.2638

Second Ring	159.413	MIN	6.2761
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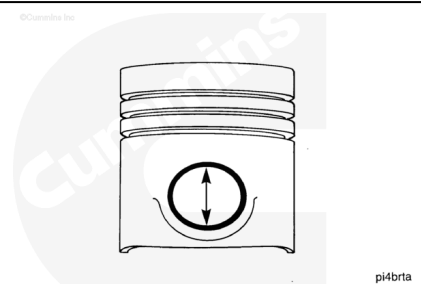
Measure the oil ring groove. The piston **must** be replaced if it is **not** within specification.

Piston Oil Ring Groove Width		
mm		in
4.788	MIN	0.1885
4.851	MAX	0.1910



Measure the piston pin bore inside diameter.

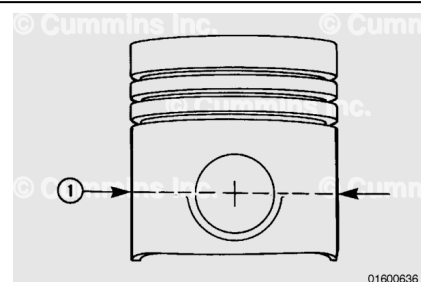
Piston Pin Bore Inside Diameter		
mm		in
60.97	MIN	2.400
61.01	MAX	2.402



Check each piston for scratches or indications of scuffing. Reference Part Reuse Guidelines, Bulletin 3810303, for details.

The piston measurement is to be recorded at room temperature of approximately 21°C [70°F].

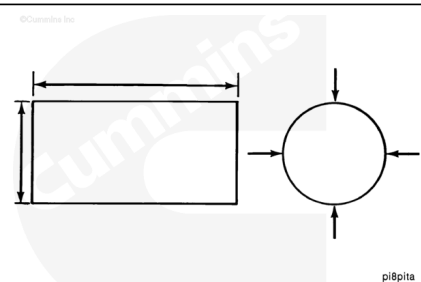
Piston Skirt Outside Diameter (1)		
mm		in
158.63	MIN	6.245
158.67	MAX	6.246



Note : Pistons are to be measured directly in line with the center of the piston pin bore and 90 degrees away from the piston pin bore.

Measure the piston pin outside diameter and length.

Piston Pin Outside Diameter		
mm		in
60.927	MIN	2.3987
60.933	MAX	2.3989



Piston Pin Length		
mm		in
118.55	MIN	4.667
118.80	MAX	4.677

Finishing Steps

- Install the piston rings. Refer to Procedure 001-047 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-047.html)
- Install the piston to the connecting rod. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-054-tr.html)

Last Modified: 27-Jan-2023
